Positioning the Infiniti G20
Complete Case Solution (HAC + PCA)

Context

The dataset contains evaluations from 75 respondents who rated 10 cars on 15 perceptual attributes and provided an overall preference score (1–10 scale) for each vehicle.

The analytical process includes:

- Constructing a perceptual map using PCA applied to the mean attribute ratings.
- Running hierarchical agglomerative clustering (HAC) on individual-level preference profiles to identify three market segments.
- Calculating segment-level mean preferences and projecting them as supplementary quantitative variables in the PCA.
- Using these perceptual and preference maps to answer strategic positioning questions for the Infiniti G20.

1 Running PCA with supplementary variables

2 Question 1

Question

Describe the two (or three) underlying dimensions of the perceptual maps. Based on these maps, how do consumers perceive the Infiniti G20 relative to competitors?

Answer

The PCA diagnostics indicate:

- The first two principal components explain approximately 70% of overall variance.
- The first three components together explain about 84%.

Thus, a 2D perceptual map already provides a rich and interpretable summary of the market, while a 3D solution offers more completeness but adds relatively less managerial insight.

Interpretation of the components.

Based on attribute loadings, the axes can be interpreted as follows:

- **Dimension 1 (Image / Quality / Prestige).**
  This dominant axis synthesizes high-level evaluative cues: attractiveness, quietness, high prestige, perceived success, and good build quality load strongly and positively, while "unreliable" and "poorly built" load negatively. It separates refined, premium-feel brands (BMW, Saab, Audi) from lower-image, mainstream models. Strategically, this is the key competitive battlefield.

- **Dimension 2 (Performance vs. Practicality).**
  This axis reflects a trade-off between driving excitement and functional utility. Sporty, interesting-to-drive cars load high on this dimension, whereas roominess and comfort load in the opposite direction. It therefore opposes emotionally appealing, performance-oriented sedans to more spacious, family-oriented vehicles.
- **Dimension 3 (Economy / Value).**
  This dimension captures the value-for-money and operating-cost perspective. “Economical” and “poor value” contribute in opposite directions, summarizing how consumers weigh cost of ownership against perceived benefits. It becomes more important when comparing mainstream brands or when budget constraints dominate decisions.

```
##        eigenvalue percentage of variance cumulative percentage of variance
## comp 1 7.74512455             51.6341637                          51.63416
## comp 2 2.79530611             18.6353741                          70.26954
## comp 3 2.06245182             13.7496788                          84.01922
## comp 4 1.27649689              8.5099792                          92.52920
## comp 5 0.44215283              2.9476855                          95.47688
## comp 6 0.38774385              2.5849590                          98.06184
## comp 7 0.20144081              1.3429387                          99.40478
## comp 8 0.06889600              0.4593067                          99.86409
## comp 9 0.02038714              0.1359143                         100.00000
```

Perceived position of the Infiniti G20.

Across the maps, the Infiniti G20 consistently clusters with compact European sedans such as the BMW 318i and Saab 900. It lies in the high-image region of the space: consumers perceive it as refined, quiet, well built, and moderately prestigious. It is clearly distinct from lower-priced domestic or mainstream Japanese cars, which score lower on quality and prestige and are more strongly associated with practicality or economy.

Conclusion.

Consumers perceive the G20 as a refined compact sedan with a strong European orientation. Infiniti’s intended positioning as a “Japanese car with a German feel” is therefore broadly confirmed by the perceptual data: the product delivers the promised sensory and image experience, even if this does not automatically translate into top preference.

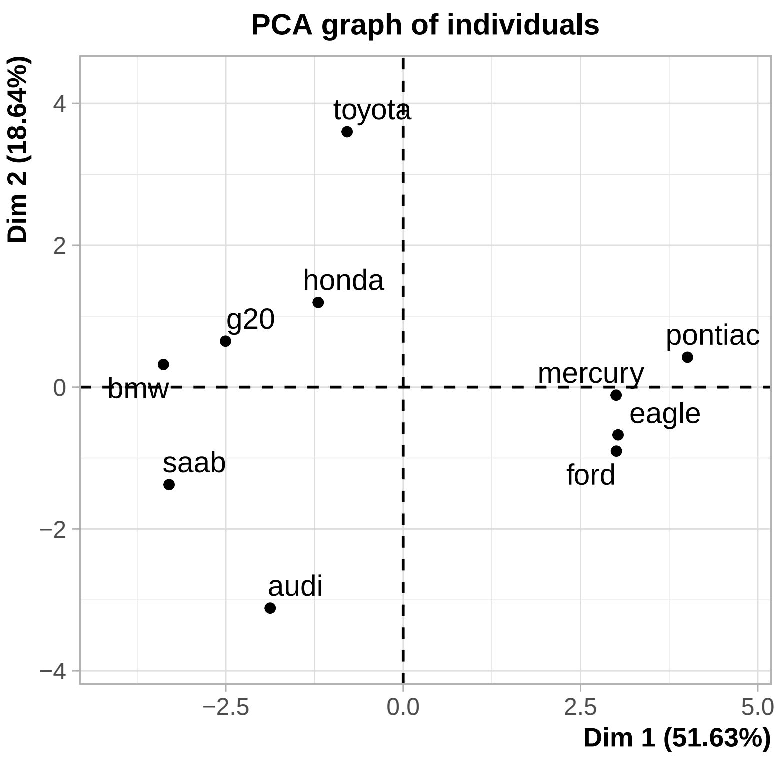

Figure 1: PCA individuals map (cars on Dimensions 1 and 2)

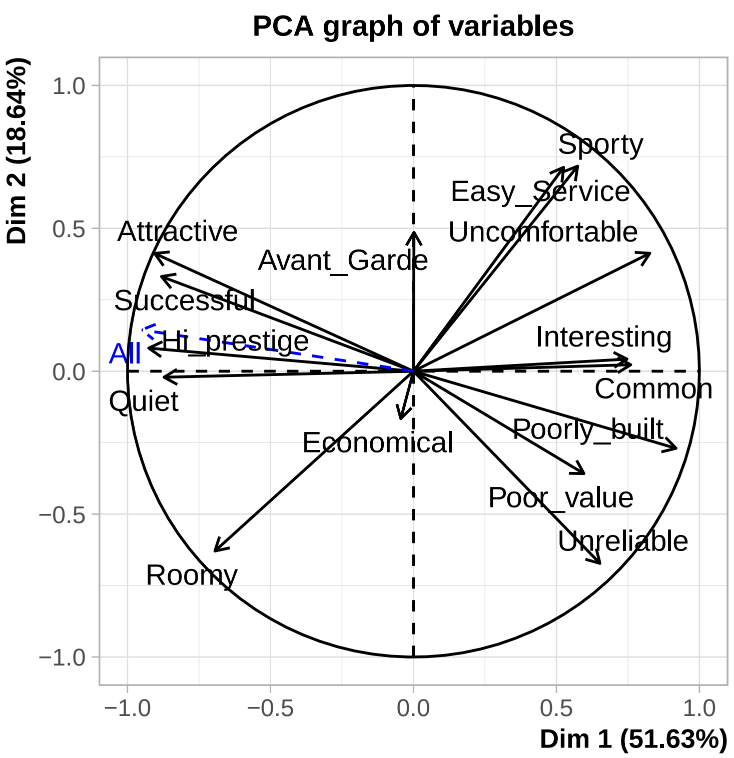

Figure 2: PCA variables map (attributes on Dimensions 1 and 2)

3 Question 2

Question

Is Infiniti's claim—that the G20 is similar to the BMW 318i but at a lower price—credible based on the perceptual and preference maps? Would explicitly accounting for price change this assessment?

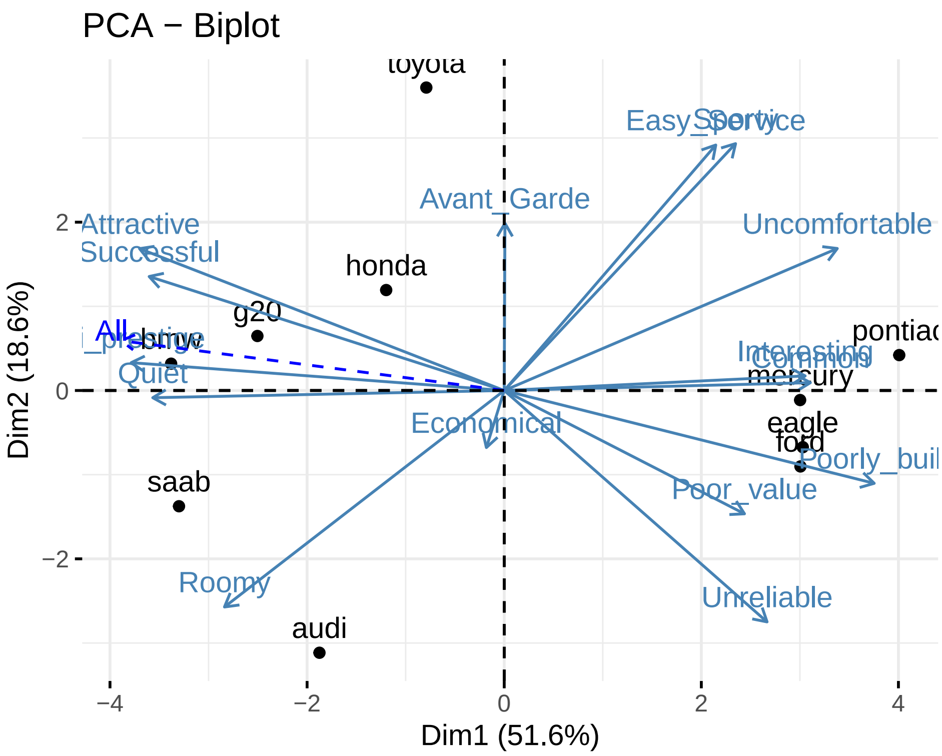

Figure 3: Biplot of PCA individuals and variables

Answer

Perceptions.
On the perceptual map the G20 is positioned within the European competitive set and lies close to BMW 318i and Saab 900. It is clearly differentiated from lower-end domestic and mainstream Japanese models. Thus, the "German feel" component of Infiniti's positioning statement is credible in perceptual terms.

Preferences.
Preference data, however, tell a different story:

- In the full sample, BMW 318i and Saab 900 are preferred to the G20 on average.
- Within each of the three identified segments, the G20 rarely emerges as the top individual choice.

In other words, the G20 is perceived as similar to BMW, but not sufficiently compelling to become the most-preferred option. Similarity in perceptual space is necessary, but not sufficient, to drive preference. **Would price correction change the conclusion?**

To evaluate whether Infiniti's lower price could realistically reverse this ranking, we introduce price in a way that is conceptually simple and geometrically consistent.

1. First, we interpret preferences as a *direction* in the perceptual space. The "All" segment defines an aggregate preference vector $\mathbf{u}$ in the $(\text{Dim } 1, \text{Dim } 2)$ plane.

2. Each car i has a coordinate vector $\mathbf{x}_i$ in this plane. We compute its *performance score* as the orthogonal projection of $\mathbf{x}_i$ on the preference direction:

$$s_i = \text{proj}_{\mathbf{u}}(\mathbf{x}_i).$$

   The larger s_i, the more the car lies in the direction that consumers prefer, given the underlying perceptual structure.

3. We then introduce price p_i by constructing a *value score*:

$$v_i = \frac{s_i}{p_i},$$

   which can be read as "performance in the preferred direction per dollar spent".

Because the projection is a linear operation, doing "projection then dividing by price" is equivalent to rescaling the coordinates by $1/p_i$ first and then projecting:

$$\text{proj}_{\mathbf{u}}\left(\frac{\mathbf{x}_i}{p_i}\right) = \frac{1}{p_i}\,\text{proj}_{\mathbf{u}}(\mathbf{x}_i).$$

However, for interpretation, it is much clearer to describe this as (i) computing a performance score s_i from the perceptual map and (ii) converting it into a value-for-money score $v_i = s_i/p_i$.

When we compare the resulting value scores v_i for BMW 318i and Infiniti G20, we obtain the same qualitative conclusion as in the unadjusted map:

- BMW's raw performance score s_{BMW} is substantially higher than s_{G20} because it is more strongly aligned with the preference direction, especially on the dominant Image/Prestige dimension.
- Dividing by price penalizes BMW, but not enough to overturn this advantage: v_{BMW} remains higher than v_{G20}.
- Infiniti's lower price improves its value signal, but the gap in perceived prestige and image is still too large.

In other words, lower price increases the G20's value-for-money but does *not* elevate it to the top choice. BMW remains the preferred option because its superior alignment with the key image and prestige attributes more than compensates for its higher price.

Managerial conclusion.

Price correction confirms that Infiniti's positioning as a "Japanese car with a German feel" is perceptually credible, but not differentiated enough to displace BMW in preference. Pricing can strengthen the value story, but it cannot by itself compensate for a weaker prestige signal and lower brand equity.

4 Question 3

4.1 Barplot of inter-class inertia losses

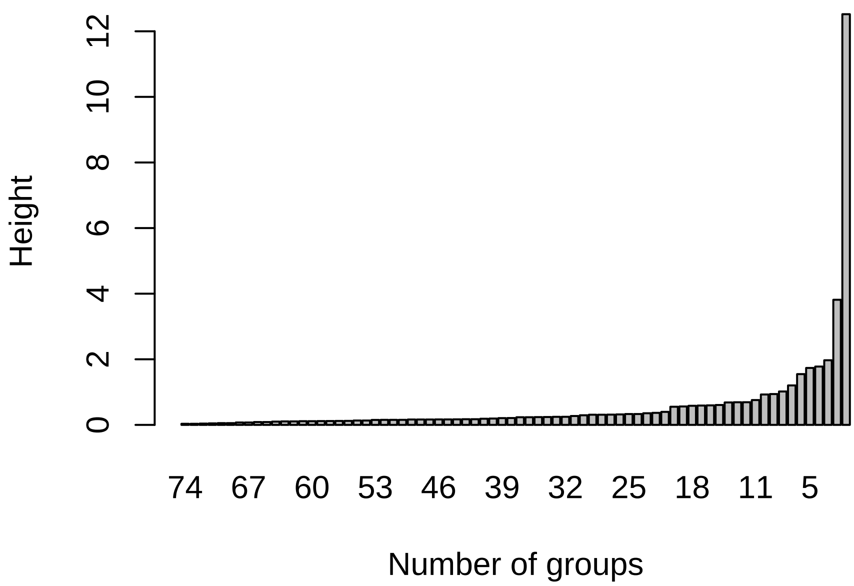

Figure 4: Inter-class inertia loss by number of groups

4.2 Dendrogram

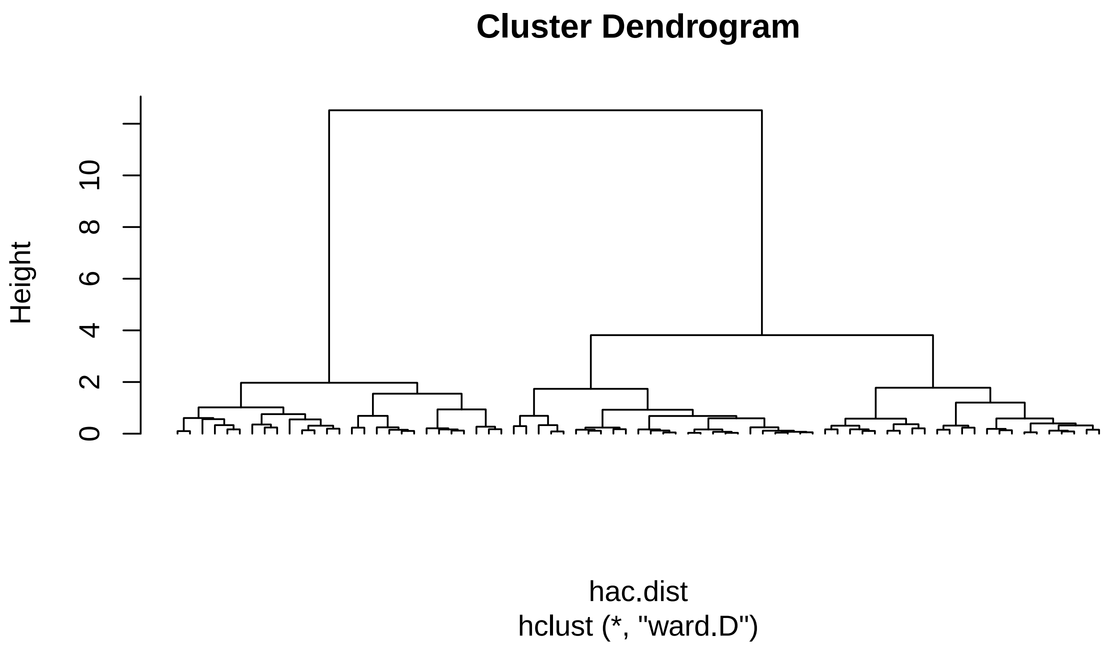

Figure 5: Cluster dendrogram (Ward's method, squared Euclidean distances)

4.3 Segment definitions used in PCA

```
##           g20     ford     audi   toyota    eagle    honda     saab  pontiac
## All  6.973333 3.866667 5.520000 5.720000 4.333333 6.773333 6.960000 3.600000
## Seg1 5.888889 5.777778 7.814815 3.518519 3.370370 5.518519 5.518519 2.592593
## Seg2 6.956522 3.782609 5.260870 6.478261 3.565217 6.608696 6.956522 3.652174
## Seg3 8.160000 1.880000 3.280000 7.400000 6.080000 8.280000 8.520000 4.640000
##           bmw  mercury
## All  6.600000 3.893333
## Seg1 5.629630 4.777778
## Seg2 6.521739 3.260870
## Seg3 7.720000 3.520000
```

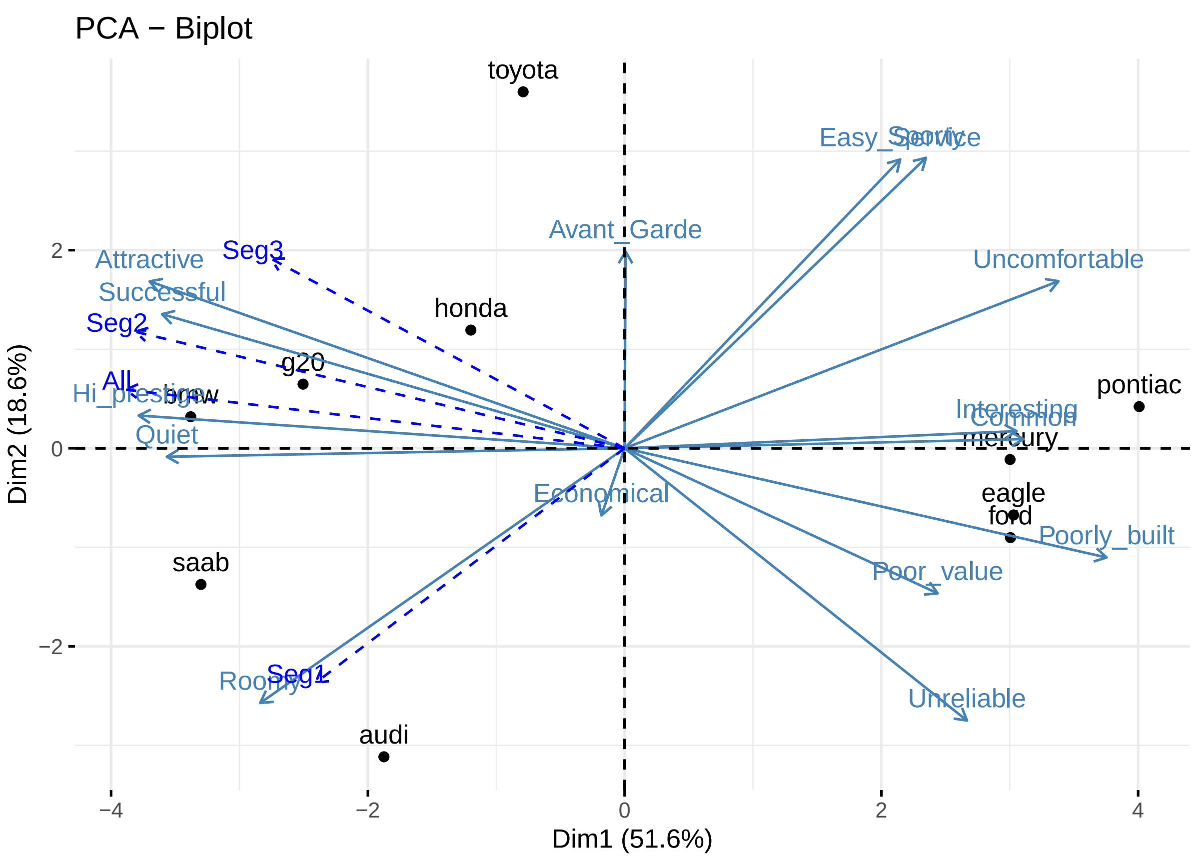

Figure 6: Enhanced PCA biplot with supplementary segment variables

Answer

Attribute drivers by segment.

Projecting segment-level mean preferences as supplementary variables reveals distinct motivational patterns:

- **Segment S1 – Functional Seekers.**
  Values roominess, comfort, quietness and reliability. This segment resembles larger households looking for practical, spacious and dependable vehicles. It aligns strongly with the Audi 90. For S1, the G20 is perceived as too compact; it does not deliver the interior space they prioritize.

- **Segment S2 – Prestige/Status Seekers.**
  Heavily driven by prestige, status and overall image. BMW 318i and Saab 900 are the clear reference brands, closely associated with high-prestige and success attributes. The G20 is positioned in the right perceptual zone but lacks sufficient prestige intensity to displace these incumbents.
- **Segment S3 – Style/Attractiveness Seekers.**
  Emphasizes attractiveness, styling and interesting design. The G20 is rated favourably here and perceived as appealing, but it faces direct competition from BMW, Saab, Toyota and Honda. It is liked, but not uniquely preferred.

Key difficulty.
The core challenge for Infiniti is that the G20 is *not* the top choice in any of the segments. It is consistently a "good second"—liked, but usually behind another brand that better matches each segment's dominant motivation (roominess for S1, prestige for S2, striking style for S3).

Target segment(s) and repositioning

Given its perceptual strengths (quietness, quality, moderate prestige) and inherent limitations (compact size, limited roominess), the G20 should not be targeted at S1. Instead, it is best oriented toward:

- **Segment S2** (prestige/status seekers), and
- **Segment S3** (design/attractiveness oriented).

For these segments, the G20 already has a strong base of perception; the strategic focus should be to sharpen its position so that it becomes *the* preferred option rather than a strong runner-up.

Repositioning strategy.

Product.

- Subtle exterior restyling to enhance attractiveness and European visual cues.
- Upgraded interior materials, fit-and-finish, and tactile quality to reinforce refinement.
- Additional work on noise insulation to further differentiate on quietness.

Price.

- Maintain a price below BMW and Saab to deliver a credible "accessible premium" proposition.
- Avoid deep discounting or overly aggressive promotions that would undermine the prestige signal and reposition the G20 as a "bargain" instead of a premium alternative.

Place.

- Strengthen the Infiniti retail experience: upscale showroom ambience, attentive personalized service, and a premium test-drive process.
- Ensure test-drives allow prospects to experience refinement, quietness and handling on high-quality routes, not just city traffic.

Promotion.

- For S2: emphasize sophistication, prestige and European driving character, supported by imagery and endorsements that signal social status.
- For S3: focus on design, styling, and driving enjoyment, using more emotional and visually impactful creative executions.
- For both segments: consistently highlight the "European feel with Japanese reliability" as a unique value proposition that competitors cannot match.

The overall objective is to relocate the G20's position in the perceptual map so that it becomes the *first* choice for at least one attractive segment (S2 or S3), rather than remaining an attractive but secondary option across the board.

5 Uses, advantages, and limitations of perceptual mapping

Uses.

1. Visualize the market structure from the consumer's perspective rather than the manager's intuition.
2. Identify market segments and clusters of ideal points that indicate different needs and motivations.
3. Detect gaps and potential niches for innovation, repositioning, or line extensions.
4. Identify true competitive sets—the brands that consumers actually compare when making a choice.
5. Support product redesign or relaunch decisions by locating the current position and desired future position on the map.
6. Evaluate the impact of marketing actions (e.g., advertising, product changes) via before/after perceptual maps.
7. Provide intuitive visual communication tools that help managerial teams align on positioning strategy.

Advantages.

- Flexible: attributes can be chosen to match the specific managerial problem.
- Joint representation of perceptions (attributes) and preferences enables rich interpretation in a single map.
- Segments can be visually identified as clusters of ideal points or preference vectors.
- Consumers find it relatively easy to provide attribute ratings compared with more complex trade-off tasks.
- PCA diagnostics (contributions, loadings) support a clear, data-driven interpretation of the axes.
- Perceptual maps support high-level strategic discussions and can be generated and updated by managers with modest technical effort.

Limitations.

1. A sufficient number of brands is required to obtain a stable and interpretable map.
2. Stated preferences can differ from actual choices; results depend on the behavioural choice rule used to translate preference into share.
3. The analysis relies on the attributes selected; omitting important attributes can distort the map and understate potential new niches.
4. Perceptual mapping explains variation only along measured attributes and existing alternatives; it may not encourage truly “out-of-the-box” thinking for radically new concepts.
5. The method is descriptive rather than explanatory: it shows where brands and segments are located in perceptual space, but not the underlying psychological processes by which consumers form these perceptions and preferences.